

Core Philosophy: “AI proposes, code verifies, and humans sign off where errors are irreversible or ambiguous.”

1. ESCALATION BOUNDARY MATRIX (MIRRORING BRIEF CRITERIA 2 & 3)

Trigger / Scenario	Agent Action	Escalate?	Rationale & Boundary Defense
Column maps to 2 fields (ref)	Calls LLM for top-2 proposals (emp_id vs mgr_id)	Yes	Margin < 0.20; guessing destroys reporting hierarchy. Prompts consultant once.
High-margin column mapping	Maps autonomously (conf ≥ 0.75, margin ≥ 0.20)	No	Statistically distinct match across aliases and types; verifiable against schema.
Sensitive field mapping (Salary/DOB)	Batches candidate mappings into 1 confirmation card	Yes	High blast-radius compliance fields; one-click approval, remembered forever as rule.
Value cannot be cleaned ("LOA")	Proposes closest canonical status ("On Leave")	Yes	Unrecognized domain enum cannot be guessed; batched into 1 card, saved as rule.
Ambiguous dates (all days ≤ 12)	Holds format; prompts once per column (DD/MM vs MM/DD)	Yes	04/05/2021 is genuinely ambiguous. Blind guesses silently shift hire dates.
Date consensus (any single day > 12)	Resolves entire column format (e.g. DD/MM/YYYY)	No	A single day > 12 provides mathematical proof of column format across dataset.
Near-duplicate identity	Fuzzy name match ≥ 90 + identical DOB	Yes	Merging two distinct people into one ID is irreversible data loss.
Obvious duplicate across files	Exact match on candidate key; merges and fills nulls	No	Corroborating record identity enriches profile; non-conflicting null fills.
Sensitive conflict (Salary)	Withholds auto-merge; surfaces side-by-side diff	Yes	Discrepancy (₹80,000 HRIS vs ₹88,000 Payroll); requires human sign-off.
Non-sensitive conflict	Resolves via client precedence (hris > crm > payroll)	No	Deterministic business hierarchy; logged to reconciliation ledger for audit trail.
Record fails validation twice	Validates → Auto-fixes → Re-validates failure	Yes	Missing required PII (e.g. email) cannot be synthesized; quarantined immediately.
Whitespace, casing & phone	Strips blanks, Capitalizes Name, normalizes E.164 (+91)	No	Reversible; deterministic regex verification; zero information loss.
Target Push 422 Client Error	Halts retry; parks record; logs target 4xx rejection	Yes	Target rejects malformed payload; retrying is futile. Pre-push cards exclude 422.
Target Push 503 Transient Error	Exponential backoff with jitter (max 3 retries)	No	Downstream network/service blip; succeeds silently on subsequent attempt.

2. DELTA SOLUTIONING & PRODUCTION ARCHITECTURE

- Resolution Memory (Learned Rules Engine)
- Human resolutions (mapping, enums, date format) are fingerprinted into client-scoped rules table.
- On re-runs, learned rules fire before the escalation gate, auto-resolving repeat ambiguities.
- Pre-push review cards drop from 7 → 3 (-57%) on Run 2 on the planted fixture set.

- Audit Trail, Idempotency & Rollback
- Append-Only Event Spine: SQLite triggers forbid UPDATE/DELETE on events; guarantees an immutable audit log.
- Idempotency & Replay: Target push uses SHA256 Idempotency-Key headers; offline LLM cache uses SHA256 prompt hashing.
- Compensating Rollback: Outages trigger automated rollback (DELETE /stub/employees/{id}) to preserve target purity.

3. EMPIRICAL VERIFICATION & PRIVACY ARCHITECTURE

52 → 29	7 → 3 (-57%)	26 of 29 (90%)	88.5% → 95.1%
SOURCE ROWS → MERGED EMPLOYEES	PRE-PUSH REVIEW CARDS (PLANTED FIXTURES)	CANONICAL EMPLOYEES ZERO-TOUCH (RUN 2)	ACTION AUTONOMY SCORE (RUN 1 → RUN 2)

*Autonomy Score Definition: (Automated Decisions / Total Pipeline Actions) × 100%. On Run 2, 26 of 29 employees require zero human touch.

Enterprise Perimeter Isolation: The hosted demo runs strictly on synthetic test data with an evaluator “Reset Demo” control. In production enterprise deployments, local LLM inference (e.g. Ollama / on-prem vLLM) executes entirely inside the client VPC boundary with no row-level data leaving the client VPC. Cross-tenant synonym matching is strictly opt-in and operates only on anonymized header tokens, never row-level PII.

4. PRODUCTION ROADMAP: WHAT I WOULD BUILD NEXT

- Active Learning Feedback Loop: Cluster unmapped column tokens across enterprise tenants to suggest industry-standard synonym maps while preserving strict tenant isolation (opt-in and headers-only).
- Multi-Entity Relational Graph: Link employees to Departments, Cost Centers, and Organizational Units with topological dependency ordering during the push stage.
- Automated Schema Evolution: Detect when client source exports drift (new columns or shifted formats) and generate automated diff migration proposals.